

Unification of Relativistic Kinematics and Gravitational Dynamics from a Single Covariant Action

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Abstract

Special relativity, general relativity, and the universality of free fall are usually introduced as conceptually distinct principles: Lorentz invariance as a kinematical symmetry, Einstein gravity as a dynamical theory of spacetime, and the equivalence principle as an independent postulate. In this work we show that these elements are not logically independent.

We consider a minimal covariant action consisting of the Einstein–Hilbert term, a unit timelike vector field subject to a normalization constraint, and minimally coupled matter. Without assuming Lorentz invariance, geodesic motion, or the equivalence principle *a priori*, we demonstrate that all three arise as necessary consequences of diffeomorphism invariance and the variational structure of the theory.

We prove that local kinematics reduce to special relativity in the tangent space at every spacetime point, yielding the standard Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$. We further show that the field equations take the Einstein form with a covariantly conserved total stress–energy tensor, and that the universality of free fall follows directly from minimal coupling and the Bianchi identity. General relativity is recovered as a consistent sector of the theory, while controlled deviations are naturally parameterized by the additional timelike sector.

These results clarify the minimal structural conditions under which relativistic physics emerges and demonstrate that special-relativistic kinematics, Einstein gravity, and the weak equivalence principle can be unified within a single covariant framework, rather than postulated as independent foundations.

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HISTORICAL NOTE — Status: historical / superseded. This manuscript is retained for its rhetorical and historical value. Its central claim that special relativity, general relativity, and the weak equivalence principle “are one” / “are not logically independent” is *circular* as a foundational statement: the Einstein–Hilbert term is an explicit ingredient of the action, and the assumed Lorentzian metric signature already encodes local Minkowski structure. The paper therefore demonstrates that these principles emerge *structurally within an assumed covariant vector–tensor framework*; it does not derive them from a more primitive substrate. The corrected and canonical statement of this result is *SST-64 (Covariant Emergence, Revised)*, which explicitly separates what is *assumed* (Lorentzian metric, Einstein–Hilbert action, unit timelike field, minimal coupling) from what is *derived* (local SR kinematics, the Einstein field equations by variation, and the WEP from the Bianchi identity). In the SST canon the derivability of relativistic *kinematics* is treated as conditional (see the Relativity Emergence Ladder), while the full Einstein *dynamics* from substrate primitives remain an open research target. The word “unification” below should be read in that qualified, effective-structure sense rather than as a derivation from something more fundamental.

1 Introduction

Special relativity, general relativity, and the universality of free fall are traditionally presented as logically distinct pillars of modern physics. Special relativity is introduced as a kinematical symmetry of spacetime, general relativity as a dynamical theory of the metric, and the equivalence principle as an independent postulate relating gravity to inertia. In this work we show that these elements are not independent.

We consider the minimal covariant extension of Einstein gravity obtained by augmenting the metric $g_{\mu\nu}$ with a unit timelike vector field u^μ subject to the normalization constraint $u^\mu u_\mu = -1$, coupled to matter only through the metric. Without assuming Lorentz invariance, geodesic motion, or the equivalence principle as axioms, we demonstrate that all three emerge as necessary consequences of a single variational principle. While related structures have appeared in modified gravity and preferred-frame theories, the present work emphasizes the logical unification of relativistic kinematics, gravitational dynamics, and free-fall universality within a single minimal variational framework. Related effective-field-theoretic approaches to Lorentz symmetry and gravity, including work by Kostelecký and collaborators, provide complementary perspectives on symmetry breaking and emergence at low energies.

Our main result can be stated as follows.

Theorem (Unification of Relativistic Principles). *Let $(\mathcal{M}, g_{\mu\nu})$ be a four-dimensional Lorentzian manifold with a diffeomorphism-invariant action consisting of the Einstein–Hilbert term, a covariant sector for a unit timelike field u^μ , and minimally coupled matter. Then:*

1. *Local kinematics reduce to special relativity in the tangent space of every spacetime point, yielding the standard Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$.*
2. *The gravitational field equations take the Einstein form $G_{\mu\nu} = 8\pi G T_{\mu\nu}$, up to additional covariantly conserved stress contributions from u^μ .*
3. *The universality of free fall follows from diffeomorphism invariance and minimal coupling, implying that structureless matter follows metric geodesics.*

HISTORICAL NOTE — Assumed vs. derived. The theorem is a statement *within* the assumed covariant vector–tensor framework. The Lorentzian signature and the Einstein–Hilbert action are *inputs*, not outputs. What is genuinely derived is: (i) local special-relativistic kinematics as a consequence of the assumed metric structure; (ii) the Einstein-form field equations by metric variation; and (iii) the weak equivalence principle from the Bianchi identity plus minimal coupling.

It is *not* a derivation of the Einstein field equations from substrate primitives; that remains an open target (cf. the thermodynamic, induced-gravity, and infrared-EFT routes discussed in SST-64 and the SST canon Relativity Emergence Ladder).

The derivation relies only on diffeomorphism invariance, the existence of a Lorentzian metric, the unit-timelike constraint, and minimal coupling of matter. No additional symmetry assumptions are imposed. The present work does not introduce a new phenomenological modification of gravity, but instead clarifies the minimal variational structure required to recover known relativistic physics.

Thus, special-relativistic kinematics, Einsteinian dynamics, and the weak equivalence principle arise from the same underlying structure and need not be postulated independently. The normalization of the timelike sector enforces local Lorentzian behavior, while the Bianchi identity ensures covariant conservation and geodesic motion of matter.

This result reframes relativity as a unified effective structure rather than a collection of separate assumptions. It clarifies the minimal conditions under which relativistic physics emerges and provides a natural framework for systematically classifying controlled deviations from general relativity that remain compatible with current observational constraints.

2 Emergence of Special Relativity (Equations Only)

2.1 Kinematic setup

Let $(\mathcal{M}, g_{\mu\nu})$ be a 4D Lorentzian manifold with signature $(-, +, +, +)$. Introduce a unit timelike field u^μ obeying

$$u^\mu u_\mu = -1. \quad (1)$$

Define the spatial projector orthogonal to u^μ :

$$h_{\mu\nu} \equiv g_{\mu\nu} + u_\mu u_\nu, \quad h_{\mu\nu} u^\nu = 0, \quad h^\mu{}_\alpha h^\alpha{}_\nu = h^\mu{}_\nu. \quad (2)$$

2.2 Local inertial reduction

At any point $p \in \mathcal{M}$ choose Riemann normal coordinates such that

$$g_{\mu\nu}(p) = \eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1), \quad \partial_\alpha g_{\mu\nu}(p) = 0. \quad (3)$$

Choose a local frame so that

$$u^\mu(p) = (1, 0, 0, 0), \quad u_\mu(p) = (-1, 0, 0, 0). \quad (4)$$

Then

$$h_{\mu\nu}(p) = \text{diag}(0, 1, 1, 1). \quad (5)$$

2.3 Timelike worldlines and the Lorentz factor

Let $x^\mu(\lambda)$ be a timelike curve with tangent $\dot{x}^\mu \equiv dx^\mu/d\lambda$. Define proper time τ by

$$d\tau^2 = -\frac{1}{c^2} g_{\mu\nu} dx^\mu dx^\nu. \quad (6)$$

In the local inertial frame at p (with $x^0 = ct$) this becomes

$$d\tau^2 = dt^2 - \frac{1}{c^2} d\mathbf{x}^2 = dt^2 \left(1 - \frac{v^2}{c^2}\right), \quad v^2 \equiv \left\| \frac{d\mathbf{x}}{dt} \right\|^2. \quad (7)$$

Hence

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}, \quad \gamma \equiv \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (8)$$

2.4 Energy–momentum relations

Define the four-velocity and four-momentum:

$$U^\mu \equiv \frac{dx^\mu}{d\tau}, \quad p^\mu \equiv mU^\mu. \quad (9)$$

Normalization gives

$$g_{\mu\nu}U^\mu U^\nu = -c^2, \quad g_{\mu\nu}p^\mu p^\nu = -m^2c^2. \quad (10)$$

In the local inertial frame,

$$U^\mu = \gamma(c, \mathbf{v}), \quad p^\mu = (\gamma mc, \gamma m\mathbf{v}), \quad (11)$$

and the invariant yields

$$E^2 = (pc)^2 + (mc^2)^2, \quad E \equiv p^0 c, \quad p \equiv \|\mathbf{p}\|. \quad (12)$$

3 Single-Action Construction (SR Kinematics + GR Dynamics + Universality)

3.1 Action

$$S = S_g + S_u + S_m = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} R + \mathcal{L}_u(g_{\mu\nu}, u^\mu, \nabla u) \right] + S_m[\psi, g_{\mu\nu}], \quad (13)$$

$$u^\mu u_\mu = -1, \quad (14)$$

$$\mathcal{L}_u = -c_1 (\nabla_\mu u_\nu)(\nabla^\mu u^\nu) - c_2 (\nabla_\mu u^\mu)^2 - c_3 (\nabla_\mu u_\nu)(\nabla^\nu u^\mu) - c_4 u^\mu u^\nu (\nabla_\mu u_\alpha)(\nabla_\nu u^\alpha) + \lambda (u^\mu u_\mu + 1). \quad (15)$$

The dimensionless coefficients c_i are assumed to lie within observationally allowed bounds consistent with gravitational-wave and post-Newtonian constraints [?].

Although the Lagrangian structure overlaps with Einstein–Æther and related Lorentz-violating gravity theories, the emphasis of the present work differs. Previous studies have primarily focused on phenomenological consequences or constraints associated with preferred-frame effects. Here, the central result is instead the logical unification of special-relativistic kinematics, Einsteinian dynamics, and the weak equivalence principle as necessary consequences of a single covariant variational framework.

3.2 Field equations

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(u)} \right), \quad T_{\mu\nu}^{(m)} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}, \quad T_{\mu\nu}^{(u)} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta S_u}{\delta g^{\mu\nu}}. \quad (16)$$

$$\frac{\delta S}{\delta u^\mu} = 0, \quad \frac{\delta S}{\delta \lambda} = 0 \Rightarrow u^\mu u_\mu = -1. \quad (17)$$

Unlike previous treatments that emphasize phenomenological modifications or preferred-frame effects, the present analysis focuses on the logical interdependence of relativistic kinematics, gravitational dynamics, and free-fall universality.

3.3 Noether identity and covariant conservation

$$\nabla_\mu G^{\mu\nu} \equiv 0 \Rightarrow \nabla_\mu \left(T^{(m)\mu\nu} + T^{(u)\mu\nu} \right) = 0. \quad (18)$$

If the matter action is diffeomorphism invariant and minimally coupled,

$$\nabla_\mu T^{(m)\mu\nu} = 0 \Rightarrow \nabla_\mu T^{(u)\mu\nu} = 0. \quad (19)$$

3.4 Local SR limit

As shown in Sec. ??, the local inertial limit yields the standard special-relativistic kinematics. Consistency with the observed equality of gravitational-wave and electromagnetic signal speeds further constrains combinations of the coefficients c_i , notably requiring $c_1 + c_3 \approx 0$. Additional bounds arise from stability, causality, and post-Newtonian analyses (see, e.g., [?] and references therein), which collectively restrict the viable parameter space without affecting the structural results derived here.

3.5 Universality of free fall (geodesic limit from minimal coupling)

For a structureless point particle with minimal coupling,

$$S_{\text{pp}} = -mc \int d\tau = -mc \int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}, \quad (20)$$

variation yields

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (21)$$

Equivalently, for dust $T_{\mu\nu}^{(m)} = \rho U_\mu U_\nu$ with $U^\mu U_\mu = -c^2$,

$$\nabla_\mu T^{(m)\mu\nu} = 0 \Rightarrow U^\mu \nabla_\mu U^\nu = 0. \quad (22)$$

3.6 GR recovery as a sector

If u^μ is in a configuration such that

$$T_{\mu\nu}^{(u)} = 0, \quad (23)$$

then

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}^{(m)}, \quad (24)$$

and in vacuum

$$G_{\mu\nu} = 0. \quad (25)$$

3.7 Weak-field (Newtonian) limit

Let

$$g_{00} = -\left(1 + \frac{2\Phi}{c^2}\right), \quad g_{0i} = 0, \quad g_{ij} = \delta_{ij} \left(1 - \frac{2\Phi}{c^2}\right), \quad \left|\frac{\Phi}{c^2}\right| \ll 1, \quad (26)$$

and take nonrelativistic matter $T_{00}^{(m)} \approx \rho c^2$. Then to leading order

$$\nabla^2 \Phi = 4\pi G \rho \quad (\text{when } T_{\mu\nu}^{(u)} \text{ contributes negligibly in this limit}). \quad (27)$$

4 Outlook and Discussion: Possible Departures from General Relativity

The construction presented in this work is intentionally conservative: by design it reproduces special-relativistic kinematics, Einstein gravity, and the universality of free fall in all experimentally tested regimes. Nevertheless, the framework admits controlled departures from general relativity, which become relevant only when additional dynamical structure in the timelike sector is activated. We briefly outline these possibilities without committing to a specific microscopic interpretation.

4.1 High-curvature and strong-field regimes

While general relativity is recovered whenever the stress–energy contribution of the timelike sector vanishes or remains subdominant, deviations may arise in regions of large curvature where gradients of the unit vector field become significant. In such regimes, the effective field equations take the form

$$G_{\mu\nu} = 8\pi G \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(u)} \right), \quad (28)$$

with $T_{\mu\nu}^{(u)}$ no longer negligible. Observable consequences may include small corrections to:

- the innermost stable circular orbit (ISCO) around compact objects,
- strong-field lensing and time-delay observables,
- quasi-normal mode spectra of black holes.

In particular, deviations in the quasi-normal mode spectrum or ISCO location provide natural observational targets, as these quantities are increasingly accessible through gravitational-wave observations and high-resolution astrophysical measurements. These effects are suppressed in weak fields and therefore compatible with solar-system and binary-pulsar tests.

4.2 Preferred-time effects beyond the local limit

Although local Lorentz invariance is recovered in the tangent space at each spacetime point, global effects associated with the timelike sector may appear on large scales or in rapidly evolving backgrounds. Such effects do not violate local special relativity, but may manifest as:

$$g_{\mu\nu}^{\text{eff}} = g_{\mu\nu} + \alpha u_\mu u_\nu, \quad (29)$$

with α dynamically suppressed in ordinary conditions. Potential observational windows include cosmological evolution, rotating spacetimes, or systems with large vorticity or acceleration.

4.3 Cosmological implications

In homogeneous and isotropic spacetimes, the timelike sector naturally aligns with the cosmological frame. Its stress contribution may therefore act as an effective fluid component in the Friedmann equations,

$$H^2 = \frac{8\pi G}{3} (\rho_m + \rho_u), \quad (30)$$

where ρ_u depends on the dynamics of u^μ . This opens the possibility of explaining certain large-scale phenomena—such as early-universe dynamics or late-time acceleration—without modifying local relativistic physics.

4.4 Relation to microscopic completions

The analysis presented here is strictly effective-field-theoretic. It does not assume any particular microscopic origin for the additional timelike sector. The unit timelike field u^μ should therefore be understood as an effective low-energy degree of freedom encoding a preferred temporal structure, rather than as a fundamental violation of local Lorentz invariance.

Similar structures arise in several well-studied contexts, including Einstein–Æther theories, khronometric gravity, and analogue or emergent spacetime models, where a normalized timelike direction appears as a collective or coarse-grained variable. In the present work no specific ontological interpretation is assumed; the role of u^μ is to identify the minimal covariant structure required for the emergence of relativistic kinematics, gravitational dynamics, and universal free fall. This contrasts with the traditional approach where special relativity, the weak equivalence

principle, and Einstein’s field equations are postulated independently; here, all three emerge from a single covariant action through diffeomorphism invariance and minimal coupling.

However, the minimal structure identified in this work provides a clear target for possible ultraviolet completions: any such completion must reproduce the unit-timelike constraint, diffeomorphism invariance, and minimal coupling in the infrared in order to recover observed relativistic physics.

4.5 Summary

General relativity and special relativity emerge in this framework as robust low-energy structures, enforced by covariance and variational consistency. Departures from Einstein gravity are confined to regimes where additional dynamical stresses become relevant and are therefore both controlled and potentially testable. This separation between a universal relativistic core and well-defined deviation channels provides a systematic path for exploring extensions of general relativity without sacrificing its experimentally confirmed foundations. In this sense, the usual postulates of relativistic physics may be understood as emergent consequences of a single covariant variational structure, rather than as independent foundational assumptions.

References

- [1] A. Einstein (1915), *Die Feldgleichungen der Gravitation*, Sitzungsberichte der Preussischen Akademie der Wissenschaften (Berlin), 844–847. (permalink: <https://einsteinpapers.press.princeton.edu/>)
- [2] C. W. Misner, K. S. Thorne, J. A. Wheeler (1973), *Gravitation*, W. H. Freeman. (ISBN: 978-0716703440)
- [3] T. Jacobson, D. Mattingly (2001), *Gravity with a dynamical preferred frame*, Phys. Rev. D **64**, 024028. DOI: 10.1103/PhysRevD.64.024028
- [4] T. Jacobson (2008), *Einstein-Æther gravity: a status report*, PoS QG-PH, 020. DOI: 10.22323/1.043.0020
- [5] B. P. Abbott *et al.* (LIGO/Virgo) (2017), *GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral*, Phys. Rev. Lett. **119**, 161101. DOI: 10.1103/PhysRevLett.119.161101
- [6] B. P. Abbott *et al.* (LIGO/Virgo) (2017), *Multi-messenger Observations of a Binary Neutron Star Merger*, Astrophys. J. Lett. **848**, L12. DOI: 10.3847/2041-8213/aa91c9

Appendix X: Scope, Ontology, and Claims of Swirl-String Theory (SST)

Purpose. This appendix is a global, corpus-level specification of (i) what Swirl-String Theory (SST) asserts beyond standard frameworks, (ii) which parts are intended as orthodox-compatible bridges, and (iii) which parts are falsifiable versus conjectural. It is referenced wherever an SST result might otherwise be misread as a claim about established physics.

Reading rule (reviewer-safe). Whenever an SST derivation is stated in orthodox language for accessibility, it should be interpreted as a *bridge description* unless explicitly labeled (**New Physics**) below. In particular, SST does *not* claim that standard quantum mechanics or general relativity are “wrong” in their validated regimes; rather, SST proposes an alternative underlying

ontology and mechanism that *can* reproduce known limits while predicting constrained deviations.

X.1 What SST Replaces (New Physics)

SST introduces an underlying, structured medium and replaces the following *fundamental* postulates:

1. **Fundamental time parameter.** SST proposes that “time” is not primitive but emerges from delayed self-consistency of collective dynamics. In corpus language: **time is a derived bookkeeping variable** associated with closure/consistency conditions in the medium, not a fundamental background parameter of the theory.
2. **Postulated quantization.** SST proposes that quantization is not assumed via Hilbert-space axioms but emerges from collective stiffness plus closure constraints in circulating/delayed systems. In corpus language: **quantization is a dynamical consequence** rather than an axiom.
3. **Particle ontology as fundamental.** SST treats localized “particles” as effective, topologically stable excitations (e.g., knot/loop sectors) of the medium, rather than as fundamental point objects.
4. **Fundamental status of $\hbar$.** SST treats $\hbar$ as a derived scale arising from medium parameters and circulation/topological invariants, rather than as an irreducible constant.

These items are the *identity claims* of SST and are *not* to be rewritten as standard “memory effects”, “cavity modes”, or “effective time scales” without an explicit pointer to this appendix.

X.2 What SST Reinterprets (Bridges and Mappings)

SST provides mappings that can often be stated in orthodox-compatible language. These are presented as **bridges** unless explicitly elevated to **(New Physics)**:

1. **Mass and inertia.** SST uses a medium-functional picture: inertial and gravitational mass can be expressed as functionals of energy-density localization, coherence, and topological sector labels (bridge to effective field and condensed-matter functionals).
2. **Gravity sector.** SST can be written in a covariant, vector-tensor language similar in architecture to Einstein-Æther / khronometric-style theories, yielding preferred-frame parameters in the weak-field limit (bridge to existing constrained frameworks). The existence of a preferred foliation is an SST ontology element; the *parameterized mapping* is orthodox-compatible.
3. **Gauge structure as bookkeeping.** SST interprets internal “charges” as topological sector labels and symmetry bookkeeping of multi-director swirl/foliation structure (bridge to emergent gauge structure in many-body systems).
4. **Boundary determinacy / holography.** SST advances a *Swirl Holographic Principle*: under specified conditions (e.g., incompressibility constraints plus topological sector closure), bulk configurations are determined by boundary swirl data and sector labels. This is not assumed to require spacetime curvature; when proven conditionally it is a bridge to constraint-based reconstruction, and when asserted as a principle it is **(New Physics)**.

X.3 What SST Leaves Untouched (Validated Domains)

SST explicitly preserves the following as *validated phenomenology* within their tested domains:

1. **Operational predictions of quantum mechanics** for systems where standard QM is verified, treating SST as an underlying explanation rather than a replacement of operational rules.
2. **Relativistic kinematics and empirically validated Lorentz tests** to the extent that SST parameters are chosen/forced to satisfy known constraints (e.g., speed-of-gravity bounds and preferred-frame bounds where relevant).
3. **Standard Model phenomenology** as an effective description, with SST aiming to reinterpret (not discard) observed charge assignments, spectra, and selection rules.
4. **General relativity in weak-field regimes** where its predictions are confirmed, with SST focusing on alternative microphysical ontology and constrained deviations.

X.4 Falsifiable Claims (Minimal List)

SST is committed to falsifiers. The following are intended as **falsifiable** targets (each should be paired in the main text with an explicit observable and bound):

1. **Preferred-frame effects in inertial/gravitational response.** Velocity-dependent corrections relative to a foliation frame can be mapped to PPN-like preferred-frame parameters and constrained by existing bounds; any SST parameterization that predicts deviations above current limits is ruled out.
2. **Delay-/feedback-induced mode selection thresholds.** If quantization emerges from stiffness + closure in circulating/delayed systems, then measurable thresholds in mode-locking, spectral discretization, or phase-stability scaling laws should appear in engineered mesoscopic platforms.
3. **Existence of a universal swirl speed scale distinct from c (New Physics).** SST introduces a characteristic swirl speed $\|\mathbf{v}_0\|$ as a universal scale entering mass normalization / circulation dynamics. This is falsifiable if it implies parameter-independent scalings across disparate platforms.
4. **Achiral topological defects as dark matter candidates (New Physics).** SST proposes dark matter as electromagnetically quiet, achiral defects with specific mass/interaction signatures. This is falsifiable via astrophysical constraints and (where applicable) laboratory null tests tied to predicted couplings.

Implementation rule. Every SST paper should list (at minimum) one falsifier relevant to that paper's sector, even if it is a null-test or a parameter-bound.

X.5 Conjectural Elements (Explicitly Labeled)

The following are acknowledged as **conjectural** (programmatic, not established), and should be labeled as such wherever they appear:

1. The ultimate origin and cosmological initial conditions of the medium/foliation structure.
2. A complete, nonperturbative classification of stable knot/defect sectors corresponding to all observed particle families.

3. Any claim of full unification (time–gravity–mass) beyond demonstrated limits, until a closed derivation and constraint-satisfying parameter fit are provided.
4. Any global holography claim beyond conditionally proven boundary determinacy statements.

X.6 One-Sentence Takeaway (Corpus Rule)

Patch where SST overclaims, not where it innovates — and explicitly flag the innovation rather than hiding it inside derivations.

X.7 Citation anchors (non-exhaustive)

The following references are cited as *comparative anchors* (not endorsements of SST’s ontology): emergent QM programs, vector-tensor preferred-frame frameworks, holography in quantum gravity, and topological/solitonic matter analogs.

References

- [1] T. Jacobson and D. Mattingly, 2001, *Gravity with a dynamical preferred frame*, Phys. Rev. D **64**, 024028. doi:10.1103/PhysRevD.64.024028.
- [2] T. Jacobson, 2008, *Einstein-Æther gravity: a status report*, PoS QG-PH, 020. doi:10.22323/1.043.0020. (Also: arXiv:0801.1547)
- [3] D. Blas, O. Pujolàs, and S. Sibiryakov, 2011, *Models of nonrelativistic quantum gravity: The good, the bad and the healthy*, JHEP **04**, 018. doi:10.1007/JHEP04(2011)018.
- [4] B. P. Abbott *et al.* (LIGO Scientific Collaboration and Virgo Collaboration), 2017, *GW170817: Observation of gravitational waves from a binary neutron star inspiral*, Phys. Rev. Lett. **119**, 161101. doi:10.1103/PhysRevLett.119.161101.
- [5] B. P. Abbott *et al.*, 2017, *Multi-messenger observations of a binary neutron star merger*, Astrophys. J. Lett. **848**, L12. doi:10.3847/2041-8213/aa91c9.
- [6] G. ’t Hooft, 1993, *Dimensional reduction in quantum gravity*, in *Salamfestschrift*, World Scientific. (Also: arXiv:gr-qc/9310026)
- [7] L. Susskind, 1995, *The world as a hologram*, J. Math. Phys. **36**, 6377–6396. doi:10.1063/1.531249.
- [8] J. M. Maldacena, 1998, *The large N limit of superconformal field theories and supergravity*, Adv. Theor. Math. Phys. **2**, 231–252. doi:10.1023/A:1026654312961. (Also: arXiv:hep-th/9711200)
- [9] P. W. Anderson, 2003, *Basic notions of condensed matter physics* (context for emergence), Addison-Wesley. (Book; no DOI.)
- [10] D. Bohm, 1952, *A suggested interpretation of the quantum theory in terms of “hidden” variables I and II*, Phys. Rev. **85**, 166–193 and 180–193. doi:10.1103/PhysRev.85.166; doi:10.1103/PhysRev.85.180.
- [11] G. E. Volovik, 2003, *The universe in a helium droplet*, Oxford University Press. (Book; no DOI.)
- [12] N. Manton and P. Sutcliffe, 2004, *Topological solitons*, Cambridge University Press. (Book; no DOI.)